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Properties of Materials
Lecture 10

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Lecture 7: Practice Time, Definitions, Concepts and Terminology (3), Task
Exercise Seminar 1
Lecture 8: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4)
Lecture 9: Recap from Last Time, Task
Exercise Seminar 2 / Lecture 10: Packing of Atoms in Solids (1), Learning Objectives, The Structure of Metals and Ceramics (1), Unit Cell Parameters, Face-Centred Cubic (fcc), Hexagonal Close-Packed (hcp), Crystal Structures - Metals, Crystal Structures - Ceramics, Concepts and Terminology (5)

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Packing of atoms in solids

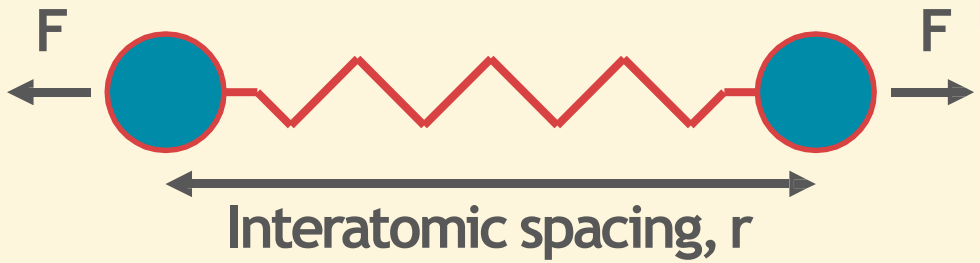
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Recap from last time...

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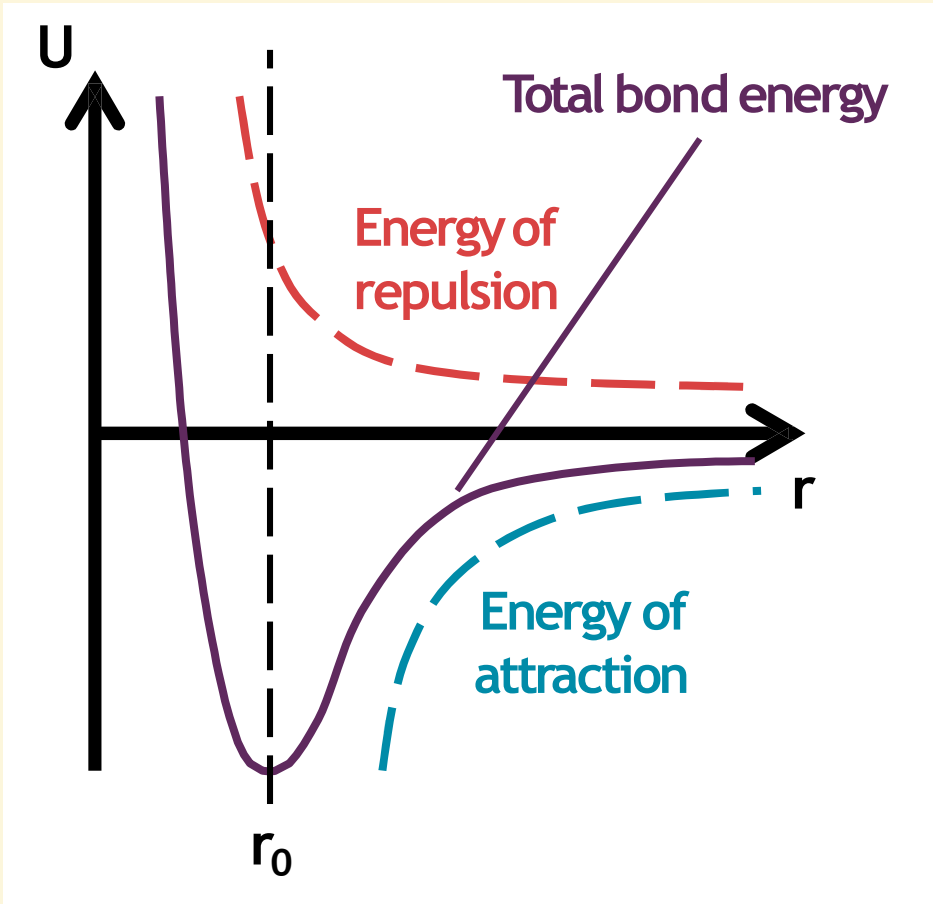
Bonding energies (1)

- Ball and spring model:
  - Atoms subjected to competing repulsive and attractive forces



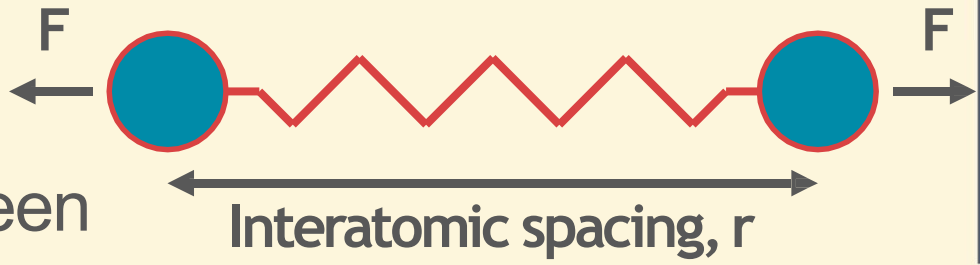
- At rest, atoms will be in their lowest energy state

- r_0 = equilibrium interatomic spacing
  U = potential energy of bond

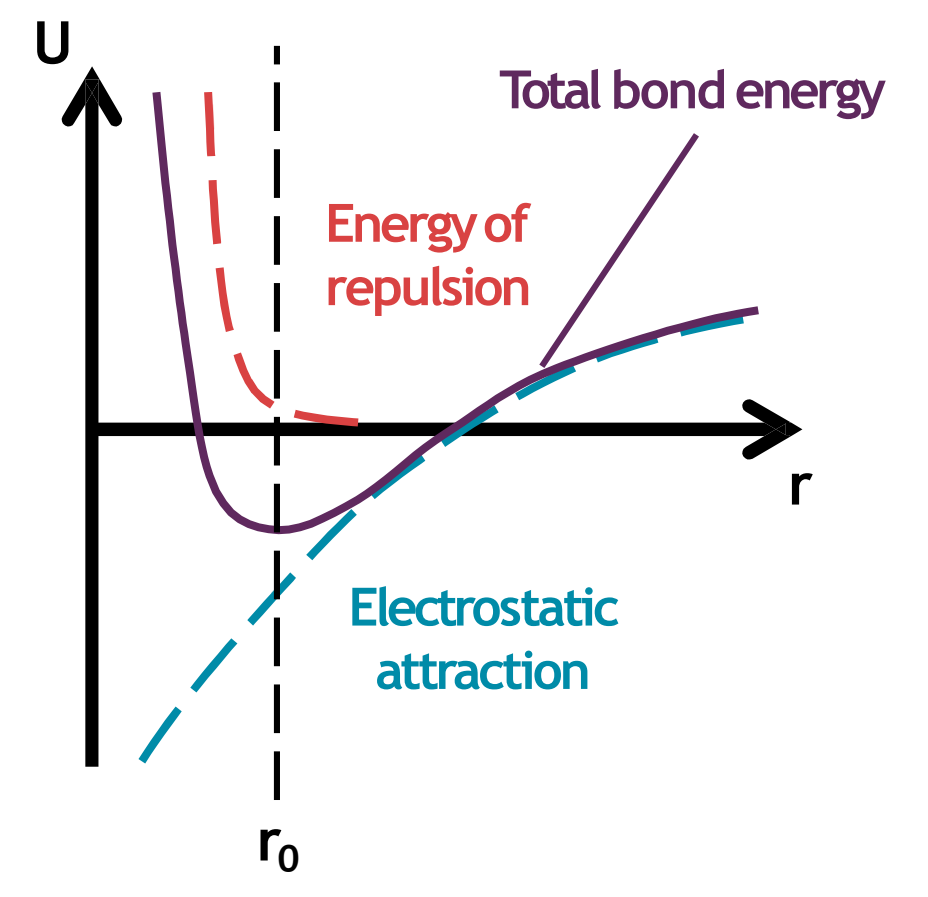


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Ionic bonding (1)

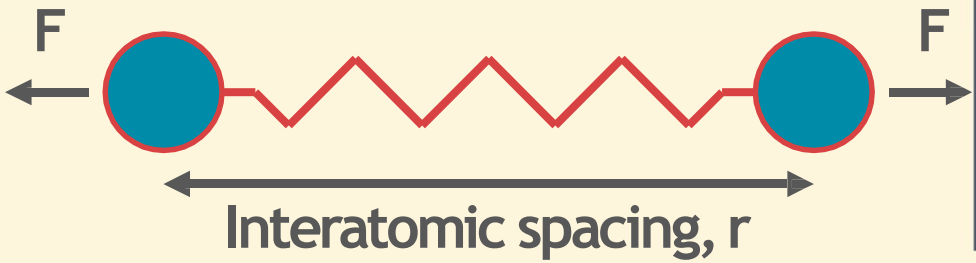


- **Electrostatic attraction** between ions:
  - e.g. $Na^+ + Cl^- \rightarrow NaCl$
- At equilibrium spacing, electrostatic attraction balanced with **repulsion energy**:
  - Ions repulse when electron charge distributions start to overlap
- Ionic bonds are **non-directional** but ionic structures must alternate from +ve to -ve

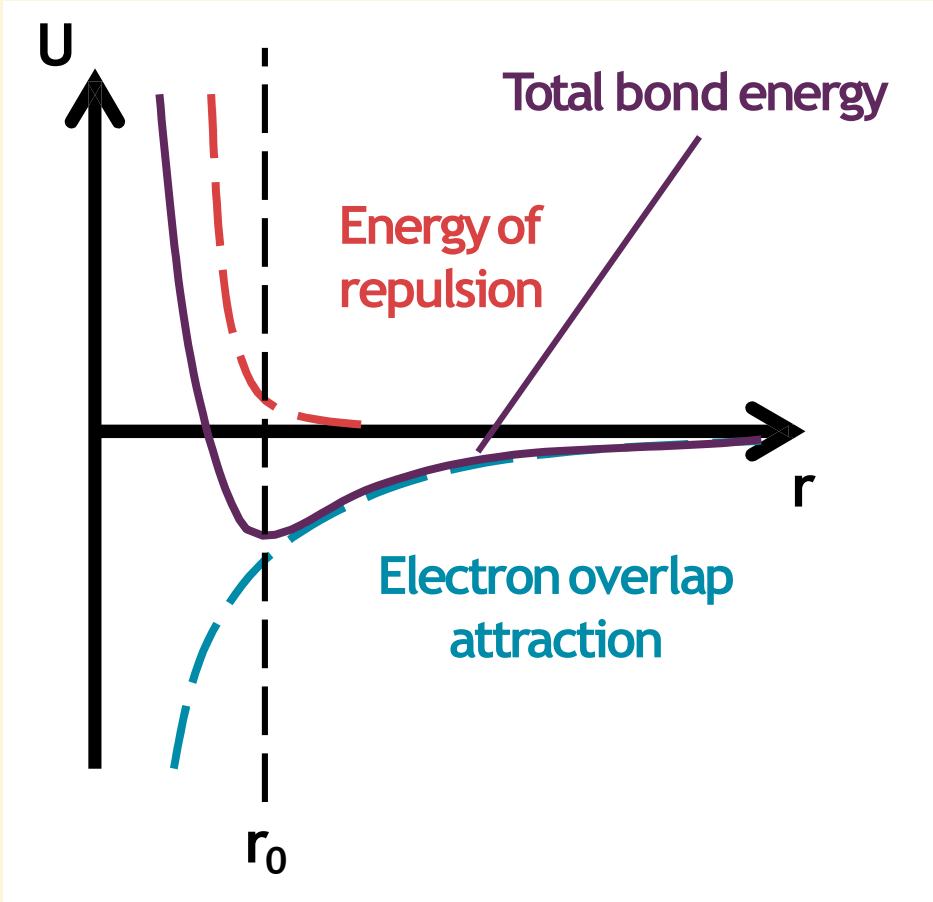


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Covalent bonding



- **Sharing of electron orbitals**:
  - e.g. Diamond, silicon carbide
  - Contributes to bonding of high-melting point metals (e.g. W, Mo, Ta)
  - Bonding in hydrocarbon chains



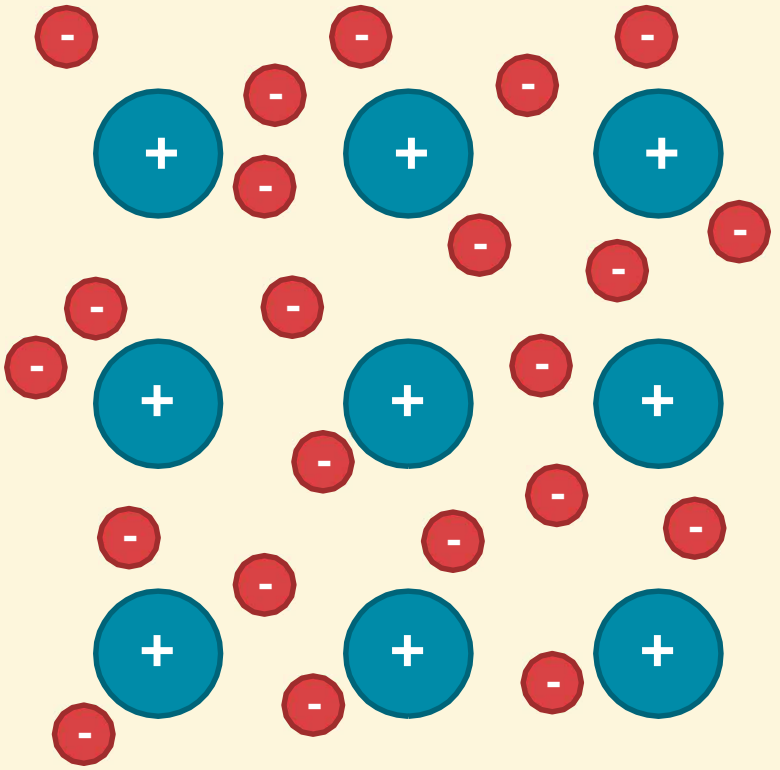
- Covalent bonds are **highly directional**:
  - Influences the way in which atoms 'stack' together to form larger structures

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Metallic bonding

- **Mobile electrons** around fixed metal cations:
  - e.g. Metals!
  - Mobile electrons give the highest electrical conductivity

- Metallic bonds are completely **non-directional**:
  - Atoms can 'stack' close together while electrons flow around them

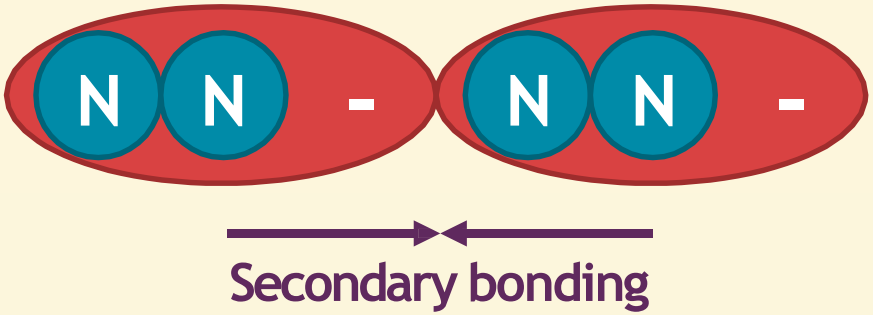


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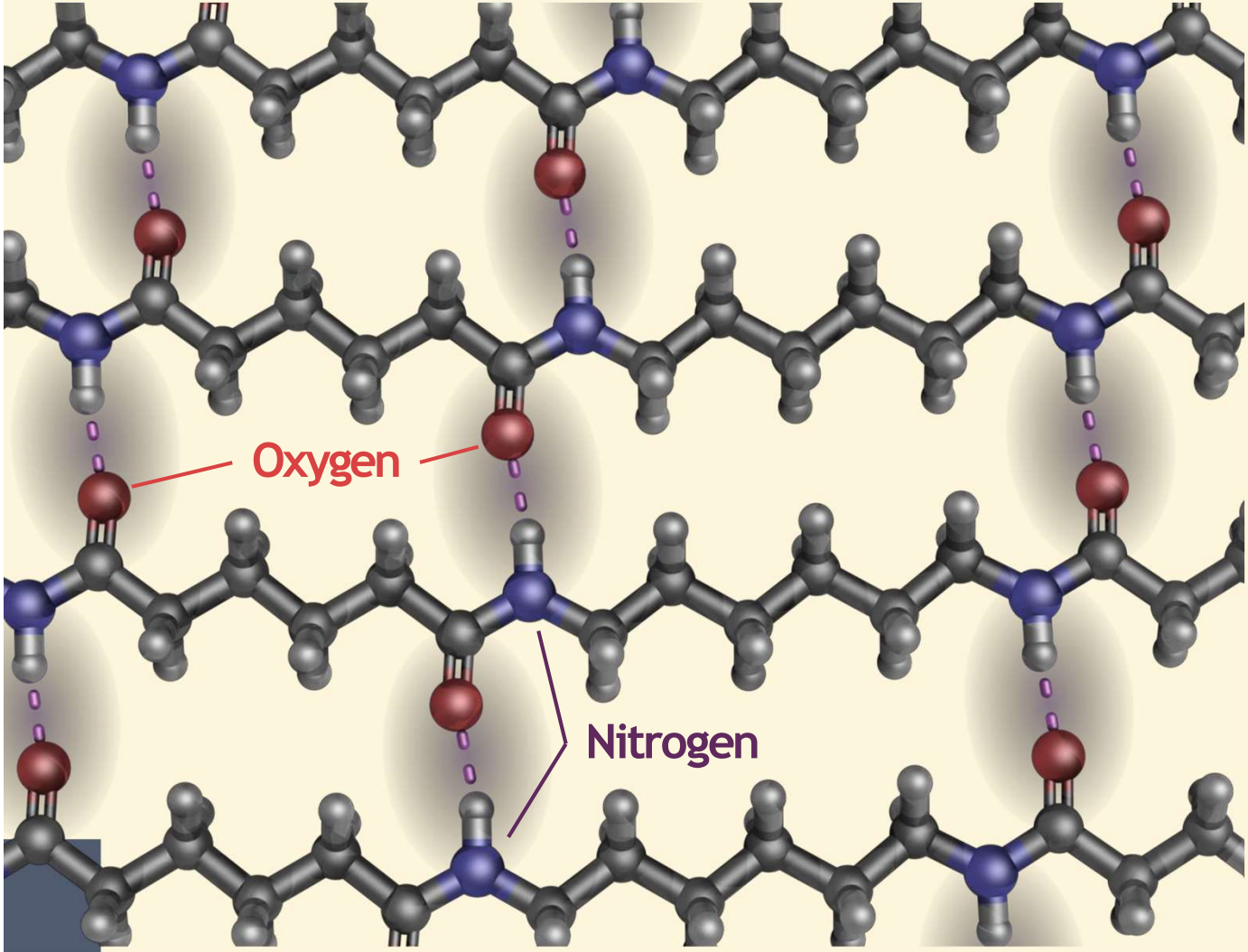
Van der Waals forces

- **Dipolar attraction** between uncharged atoms
  - Electron movement causes **fluctuation in charge** on individual atoms, which then interact to create dipoles on others

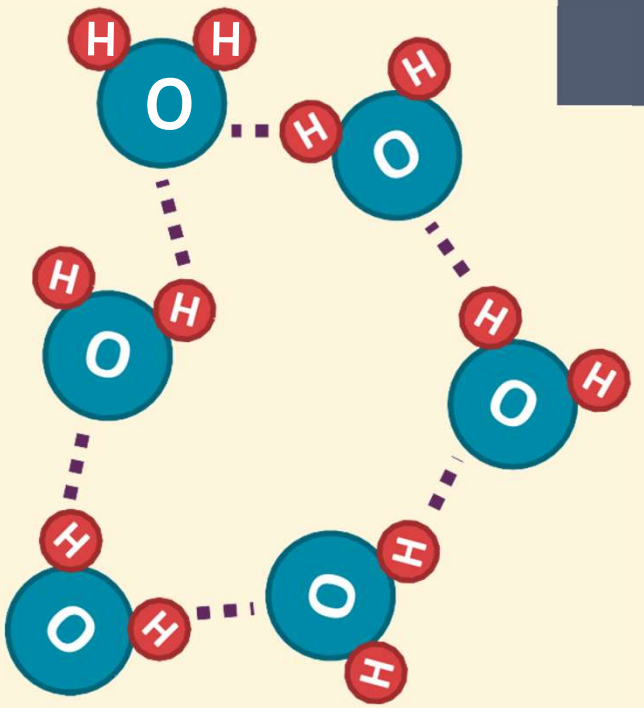
- Without Van der Waals forces, **liquefaction** would not be possible for many important industrial liquids:
  - Liquid nitrogen, oxygen, helium and hydrogen



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Hydrogen bonding in nylon6,6
https://commons.wikimedia.org/wiki/File:Nylon-3D-h_bond.png



Hydrogen bonding

- Hydrogen atoms are able to **share their charge**:
  - Only with nitrogen, oxygen or fluorine in molecules
- Essential for solid polymers and keeping water liquid!

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Bonding types and properties

- **Ceramics**:
  - Large bond energy
  - Stiff and strong, but brittle
  - High melting point and low thermal expansion coefficient
  - **Ionic** and **covalent** bonding
- **Metals**:
  - Variable bond energy
  - Moderate stiffness and strength, can be brittle or ductile
  - Tend to have high thermal and electrical conductivity
  - **Metallic** bonding
- **Polymers**:
  - Directional properties
  - Low stiffness and strength; poor thermal properties
  - **Covalent** and **secondary** bonding

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Learning objectives

- To learn the common **crystallographic structures** of metals (and some ceramics):
  - Body-centred cubic (bcc)
  - Face-centred cubic (fcc)
  - Hexagonal close-packed (hcp)

- To become familiar with **typical examples** of metals and ceramics that possess these structures

- To learn how to characterise the 'density' of different atomic arrangements by their **atomic packing factor**

Material
(10^0 m^3)

- Bulk properties:
  $E, \sigma_y, T_m, a, \lambda \ldots$

Grains / crystals
(~10^{15} per m^3)

- Microstructure: grain boundaries and precipitates
- Sub-grains, single crystal or polycrystalline

Unit cells
(>10^9 per crystal)

- Lattice structure
- Atomic bonds
- Atoms (2 – 6 per unit cell)

The structure of metals and ceramics (1)

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The structure of metals and ceramics (2)

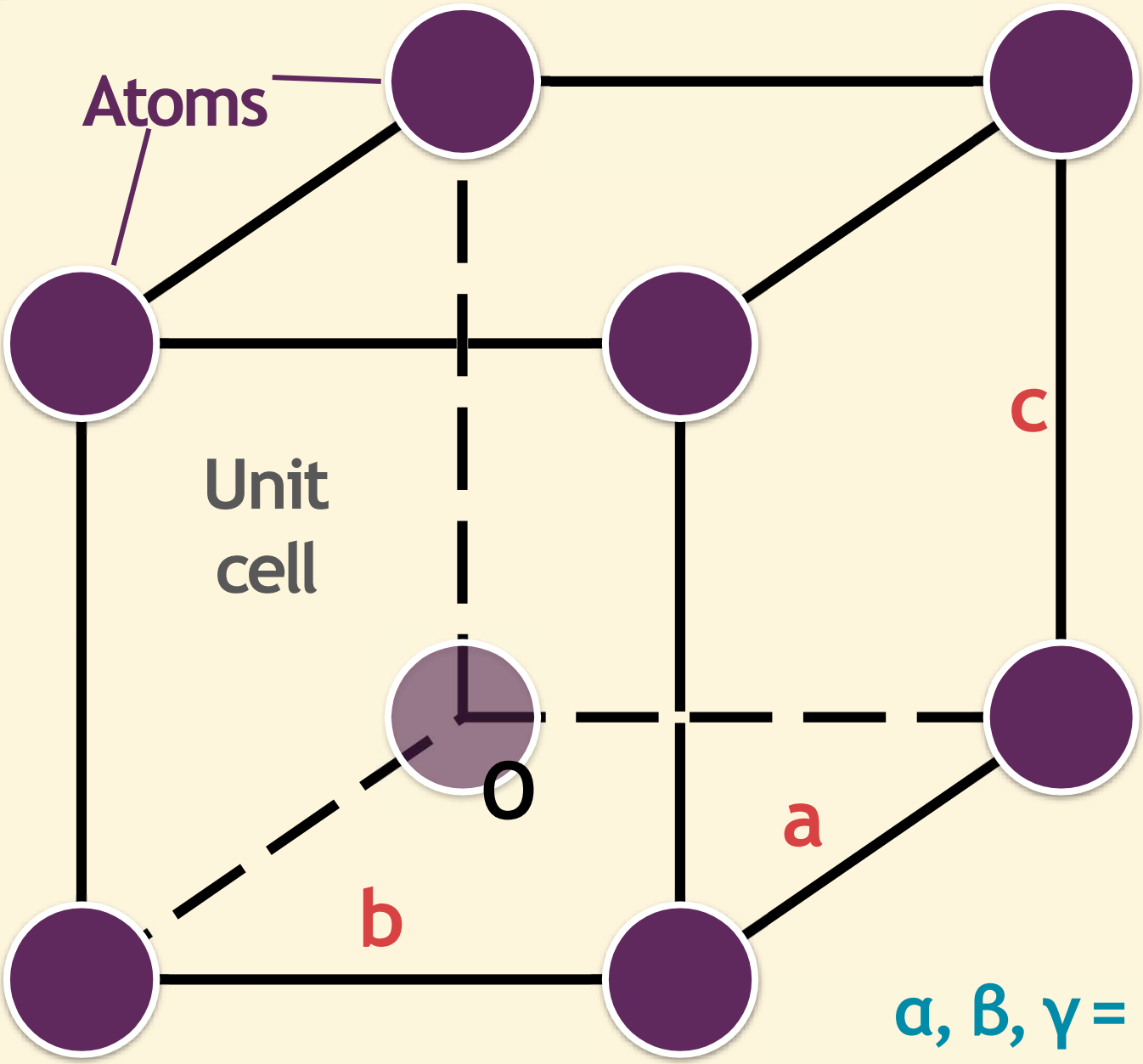
S6

S5

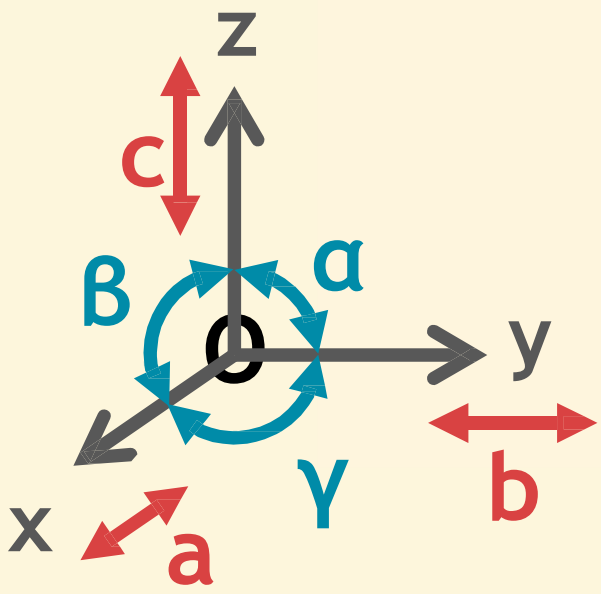
S4

- Metals/ceramics are made up of a number (~10^{15} m^{-3}) of **grains**
- *Grains are typically imperfect and are themselves typically made up of a number of **sub-grains***
- Each grain can be considered as a **crystal**; an ordered configuration of atoms
- Each crystal contains a number (>10^{9}) of **unit cells** with the same lattice structure and orientation
- Each unit cell contains a number of main / primary / **substitutional atoms** that are connected by **primary bonds**
- *Each unit cell may also contain a number of smaller secondary/ **interstitial atoms** in between larger atoms*

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Simple cubic structure

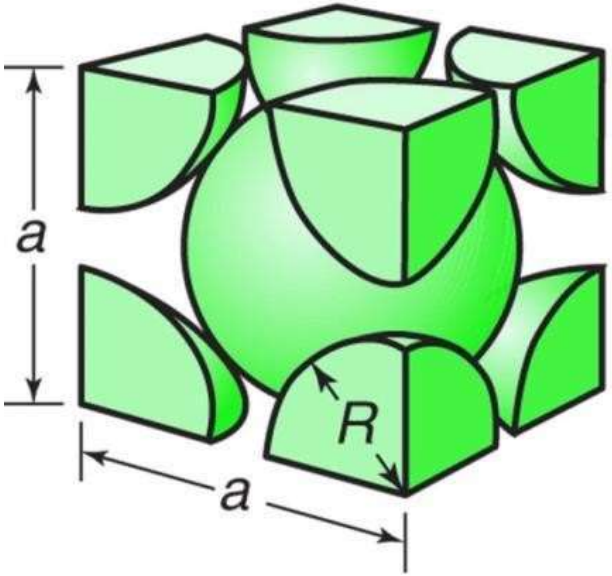


a, b, c = lattice parameters

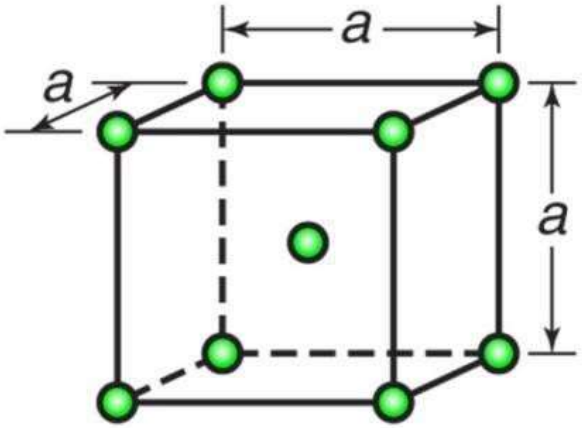
α, β, γ = lattice angles

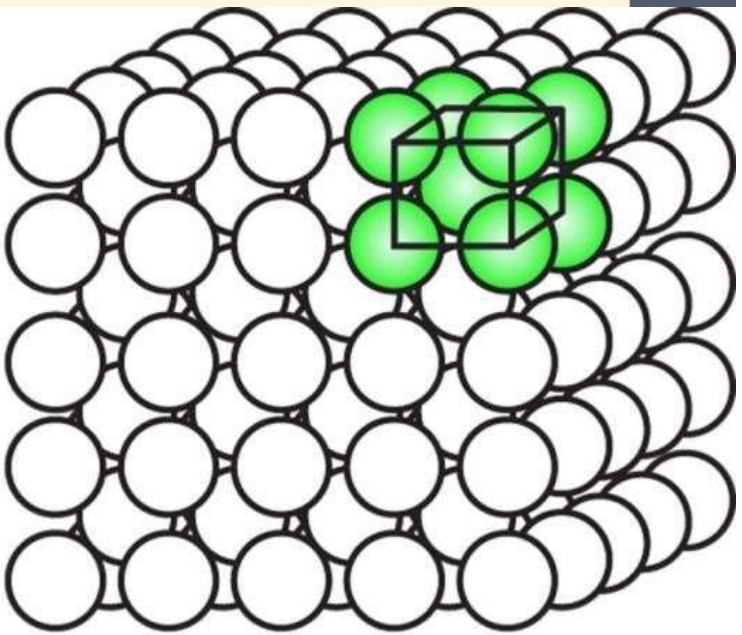
Unit cell parameters

R = atomic radius



a = lattice parameter



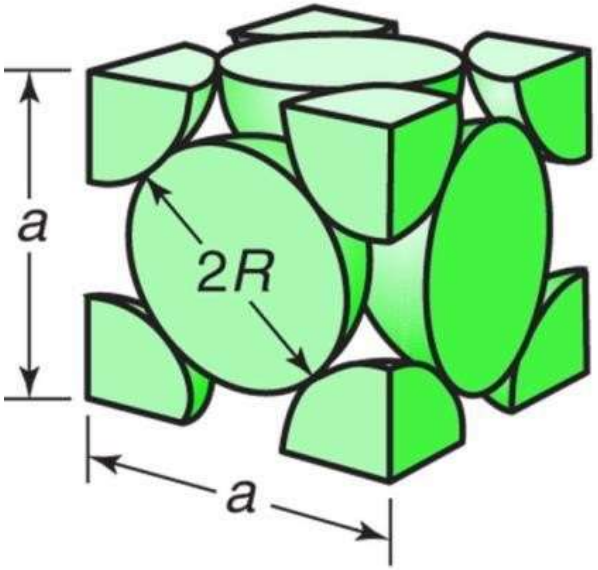


Hard-ball model	Unit cell	Single crystal – multiple unit cells

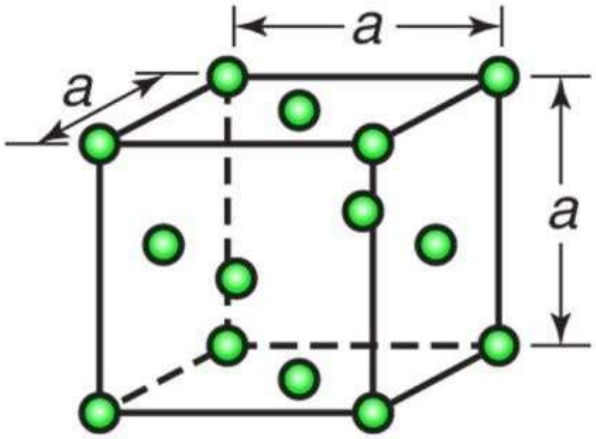
Body-centred cubic (bcc)

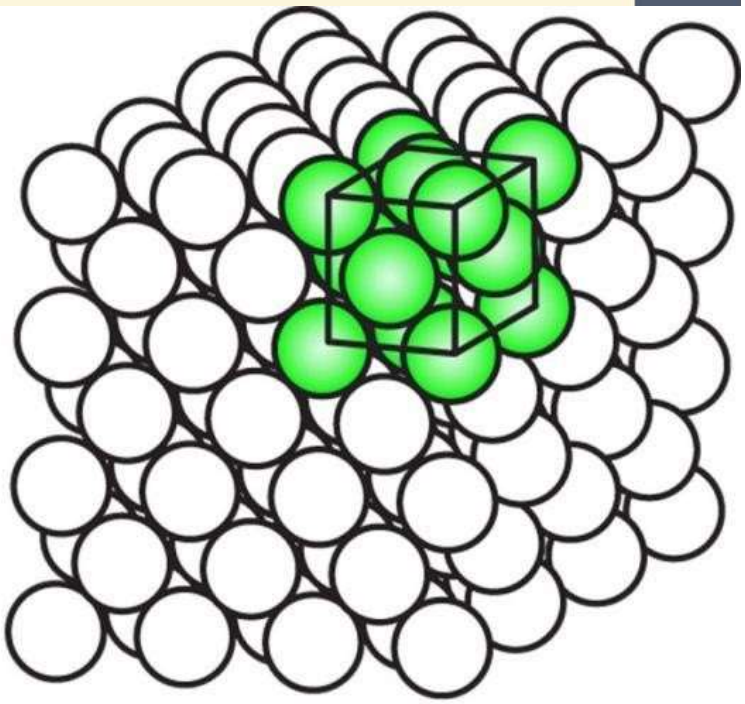
- **Unit cell**: The smallest repeating unit needed to describe the complete extended structure of a crystal (through repetition and translation)

R **= atomic radius**



a **= lattice parameter**



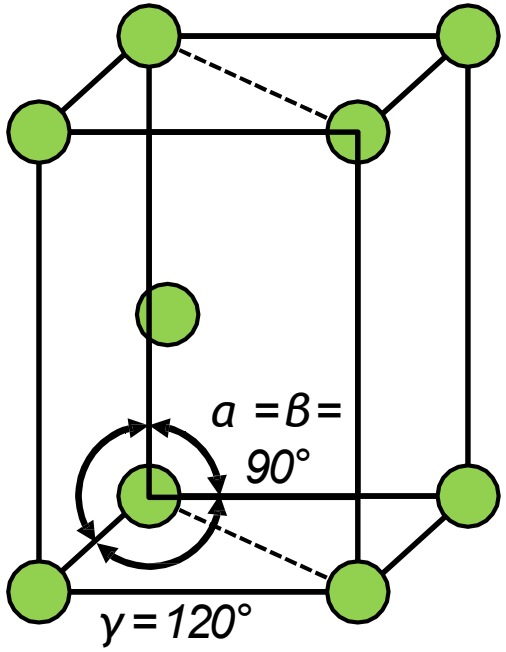


Hard-ball model	Unit cell	Single crystal – multiple unit cells

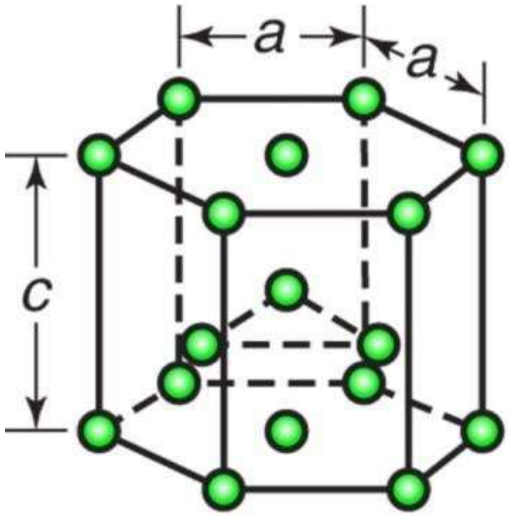
Face-centred cubic (fcc)

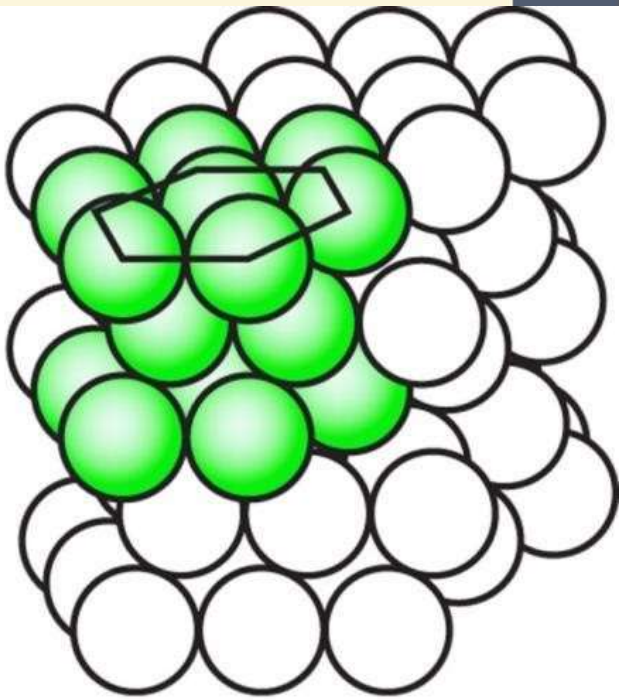
- **Close-packed**: A unit cell with the theoretical maximum **atomic packing factor** – APF $= \frac{N_{\text{atoms}} V_{\text{atoms}}}{V_{\text{unit cell}}}$

a, β, γ = lattice angles



a, c = lattice parameters





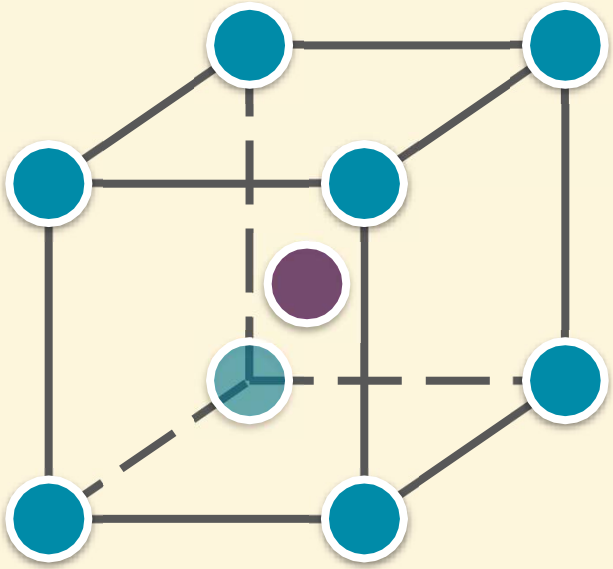
Unit cell

Single crystal – multiple unit cells

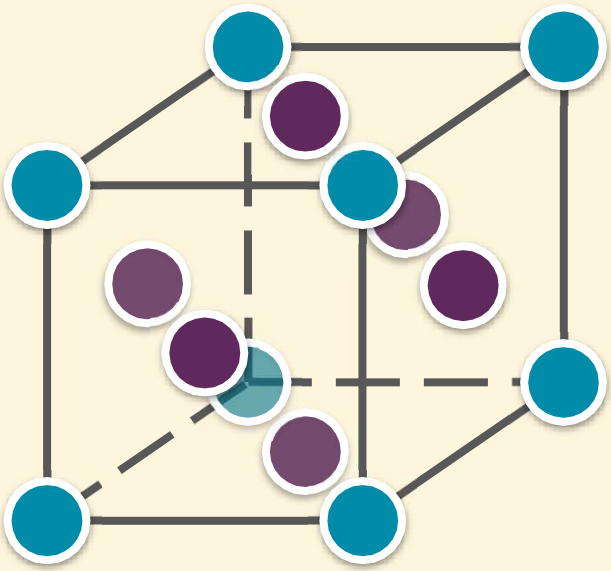
Hexagonal close-packed (hcp)

- **Close-packed**: A unit cell with the theoretical maximum **atomic packing factor** – APF = $\frac{N_{\text{atoms}} V_{\text{atoms}}}{V_{\text{unit cell}}}$

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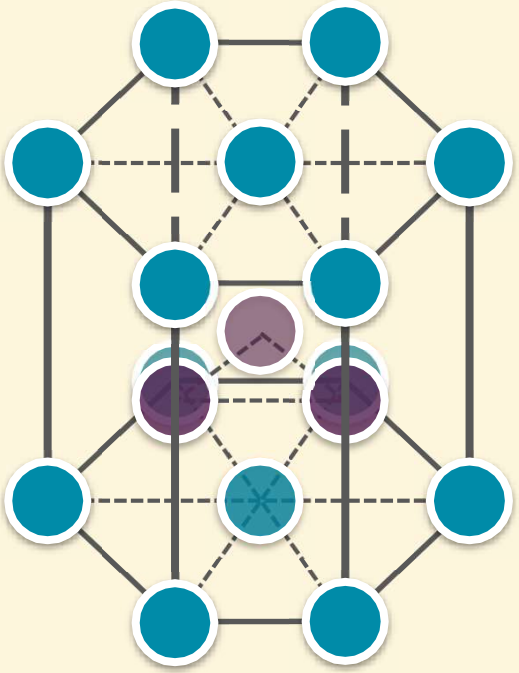


Strong / hard materials



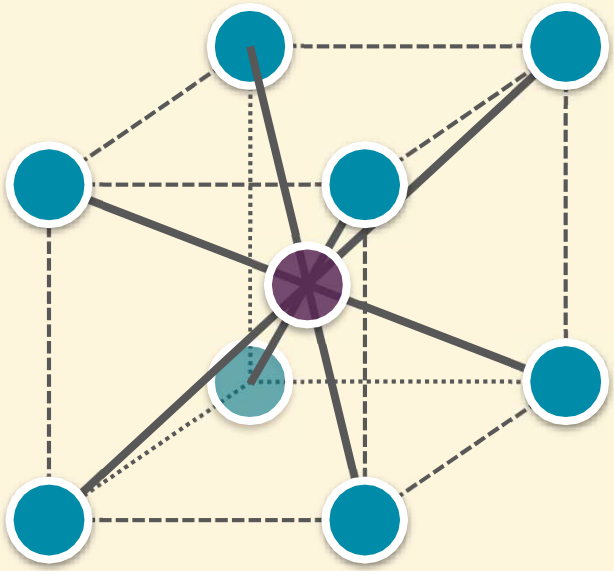
Ductile materials

(Some) light materials



Crystal structures – Metals

- **bcc**: Tungsten (W), Chromium (Cr), α-Iron (ferrite)
- **fcc**: Copper (Cu), Aluminium (Al), Nickel (Ni), γ-Iron (austenite)
- **hcp**: Zinc (Zn), Titanium (Ti), Magnesium (Mg)

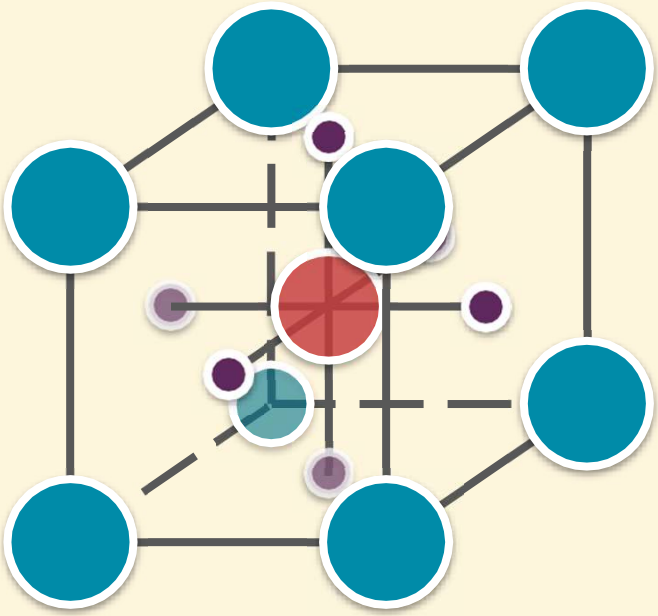


Caesium Chloride

$Cs^+ + Cl^- \longrightarrow CsCl$

Calcium Titanate

$Ca^{2+} + TiO_3^{2-} \longrightarrow CaTiO_3$



Crystal structures – Ceramics

- Ceramics have far more **complex** structures
- With a mixture of **ionic** and **covalent** bonding (high bond energies) they are very difficult to deform

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Summary

- Metals and ceramics have **crystalline structures**, formed of crystals made up of **unit cells** of **ionically**, **covalently** (in ceramics) or **metallically** (in metals) bonded **atoms**

- Many metals and ceramics have one of three crystallographic structures:
  - **Body-centred cubic** (bcc)
  - **Face-centred cubic** (fcc)
  - **Hexagonal close-packed** (hcp)

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Concepts and terminology

- Atoms:
  - Substitutional
  - Interstitial
- Unit cell
- Crystal
- Sub-grain
- Grain

- Crystallographic structure:
  - Body-centred cubic (bcc)
  - Face-centred cubic (fcc)
  - Hexagonal close-packed (hcp)